

Spectrascope

Blind analysis of raw radio captures: from IQ samples to the recovered message

Problem Statement ID	[fill in]
Problem Statement Title	[fill in from the SIH portal]
Theme / Category	[fill in]
Team ID and Team Name	[fill in]

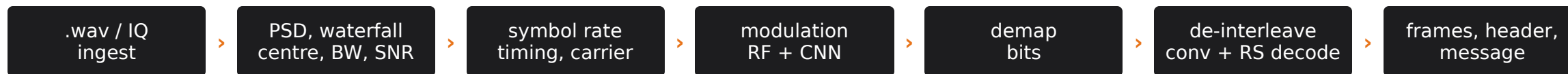
Idea

- **Proposed solution:** web tool that takes a .wav or raw IQ capture with no prior knowledge and reports sampling-derived parameters, modulation, FEC, interleaving and frame structure
- **How it addresses the problem:** replaces manual analysis with an automated chain and shows its evidence (spectrum, constellation, confidence) at every step
- **Demodulates** BPSK, QPSK, 8PSK, 16QAM, 2FSK, 4FSK
- **Decodes** convolutional codes (Viterbi), Reed-Solomon and concatenated codes; de-interleaves block, diagonal, convolutional, pseudo-random

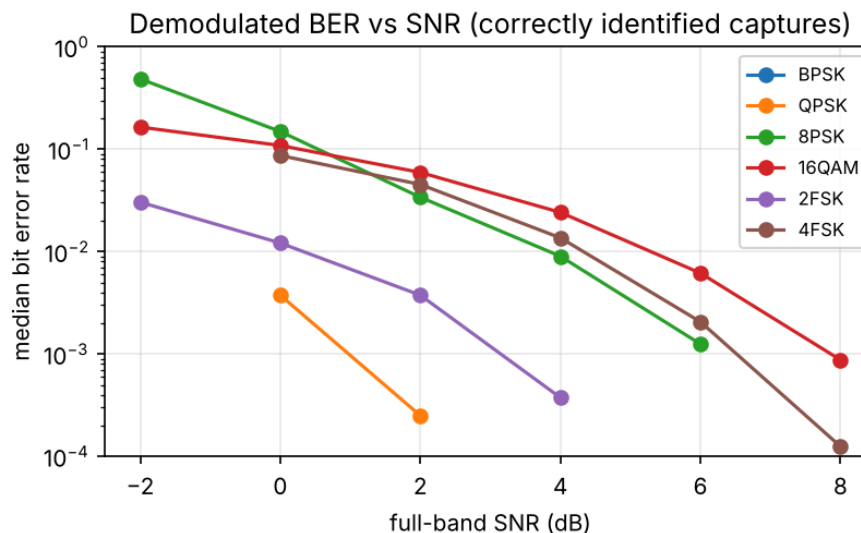
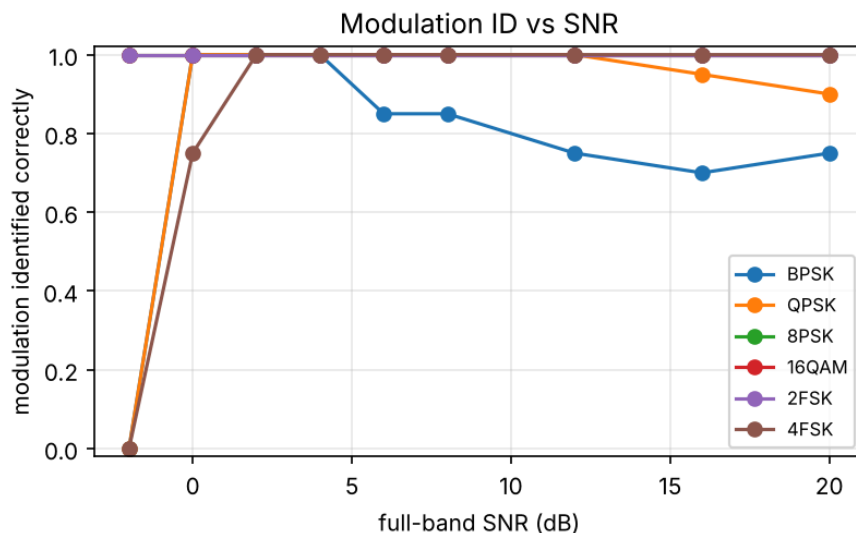
Innovation and uniqueness

- Blind identification of the convolutional code and the RS code from the bits alone
- Feed-forward estimators: no loops to tune, explainable from first principles
- Bit-stream correlation finds sync word, header and payload
- Second opinion from a CNN trained on public RadioML data
- PDF evidence report per analysis

Technical approach



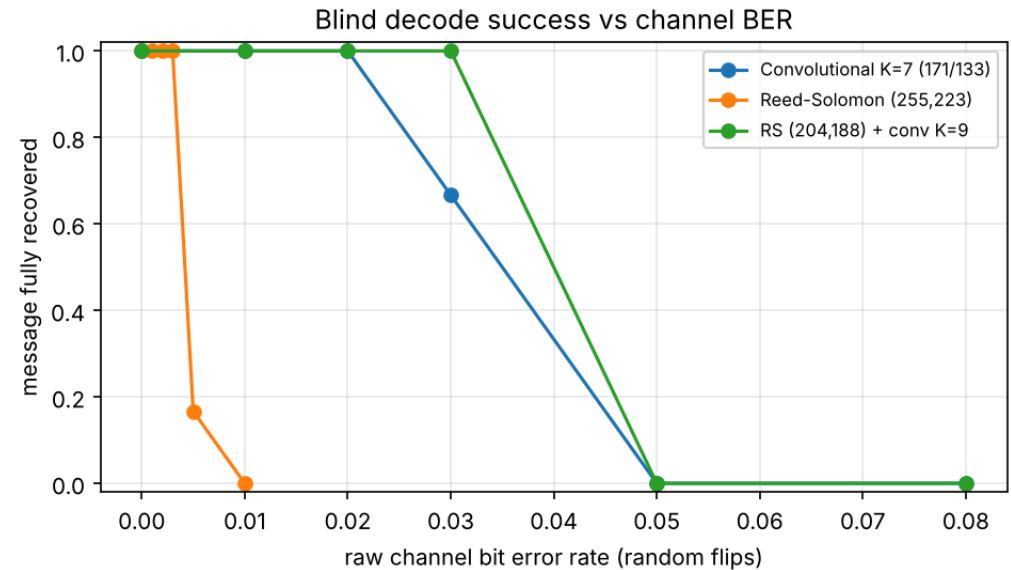
- **Technologies:** Python, NumPy/SciPy, scikit-learn, ONNX Runtime, FastAPI, plain HTML/CSS/JS | **Prototype:** working web app, 66 automated tests, PDF report export
- **Methods:** $|x|^2$ spectral line for symbol rate; Oerder-Meyr timing; M-th power carrier recovery; GF(2) correlation for code identification; Berlekamp-Massey RS decoding



Own simulator, not real captures
Own simulator, 20 captures per point. Not real captures.

Feasibility and viability

- **Feasible:** runs on a laptop CPU; CNN is a 2.3 MB ONNX file
- **Measured (simulator):** BER 0 at 8 dB full-band SNR for PSK/2FSK; conv K=7 decodes to ~2 % raw BER, RS(255,223) to ~0.3 %, RS+conv to ~3 %
- **Challenges and risks** (stated plainly):
 - No real capture tested yet; transmitter and receiver share an author
 - BPSK sometimes mistaken for FSK (15-30 % in simulation)
 - CNN agrees with truth on 71 % of simulated captures; informational only
 - No LDPC, scramblers, puncturing, soft decisions; interleaver size supplied by operator



Own simulator and own transmit chain, not real captures

Blind decode success vs raw channel BER, own simulator and transmit chain.

- **Strategy:** validate on public / RTL-SDR captures, add LDPC with supplied H, soft-decision Viterbi

Impact and benefits

- **Analysts:** minutes instead of hours for first-pass parameter identification; every conclusion shows its evidence
- **Trust:** confidence, warnings and limits are shown; nothing is presented as certain when it is an estimate
- **Training and research:** each algorithm is explainable; documented with known-answer tests
- **Reuse:** works on files from different sensors (.wav or raw IQ cf32/ci16/ci8/cu8)
- **Low cost:** open-source stack, no GPU or licence needed at run time
- **Roadmap:** LDPC, scrambler support, SigMF metadata, batch mode, GNU Radio flowgraph export

| Research and references

- O'Shea, Roy, Clancy. "Over-the-Air Deep Learning Based Radio Signal Classification". IEEE J. Sel. Topics Signal Process., 2018 (RadioML 2018.01A dataset)
- Oerder, Meyr. "Digital Filter and Square Timing Recovery". IEEE Trans. Commun., 1988
- Viterbi. "Error Bounds for Convolutional Codes and an Asymptotically Optimum Decoding Algorithm". IEEE Trans. Inf. Theory, 1967
- Reed, Solomon. "Polynomial Codes over Certain Finite Fields". J. SIAM, 1960
- CCSDS 131.0-B: TM Synchronization and Channel Coding (attached sync marker, RS(255,223), K=7 code)
- Tools: FastAPI, ONNX Runtime, scikit-learn, NumPy/SciPy

All accuracy figures: this project's own simulator. Dataset: RadioML 2018.01A (simulated).